

Multimodal Hands-Free VR For Wheelchair Users With Upper Limb Mobility Limitations: Leaning, Head-Gain, and Gaze Pointing

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ABSTRACT

The immersive experiences and social connectivity offered by Virtual Reality (VR) make it a powerful tool for promoting inclusion and accessibility. However, many VR games and applications frequently rely on body tracking or bi-manual controllers, excluding users with upper limb mobility limitations, especially wheelchair users. For these users, simultaneously managing handheld controls and manual wheelchair navigation represents a challenge and reduces engagement and the perception of accessibility. To address this, we propose a multimodal hands-free system designed for wheelchair users with upper limb mobility restrictions. Our system uses HMD-integrated sensors to perform navigation and pointing (selection) tasks in virtual environments. It combines a leaning-based locomotion system, dynamic head-gain rotation, and gaze-based pointing for precise interaction. This solution eliminates the need for additional equipment and accommodates the physical demands of these users. This paper outlines the design principles and implementation of the multimodal system, aiming to advance accessibility in VR for wheelchair users.

Index Terms: Wheelchair Users, Hands-Free, VR Accessibility, Leaning-based Locomotion, Head-Gain Rotation, Gaze Pointing.

1 INTRODUCTION

Virtual Reality (VR) gaming and social networks, such as those offered by platforms like Meta, have increased in popularity [1, 18] due to the immersive experience and ability to connect users in virtual spaces, especially through gaming [25, 35]. Despite its evolution, these experiences heavily rely on bi-manual controllers for locomotion, camera control and selection, excluding users who cannot operate such devices due to upper-limb limitations. For manual wheelchair users, this reliance is especially problematic, as it creates a conflict between operating handheld controls and manually turning the wheelchair during gameplay [9, 43, 29]. This dual-task challenge often reduces engagement and increases perceptions of inaccessibility [8, 29]. Developing hands-free interaction systems is essential to making VR accessible to this group of users [16].

Multimodal hands-free interaction techniques have been introduced to address this challenge by leveraging additional or integrated head-mounted display (HMD) sensors. For example, eye tracking, speech recognition, and head movement have been successfully combined for control in mixed reality systems [42]. Similarly, Zielasko et al. [45] proposed a torso+gaze multimodal system that separates pointing interaction from navigation tasks, enhancing user experience in navigation and selection. Sensor-based systems have adapted wheelchair interfaces to VR interactions, incorporating them as alternative input methods for games [10, 11].

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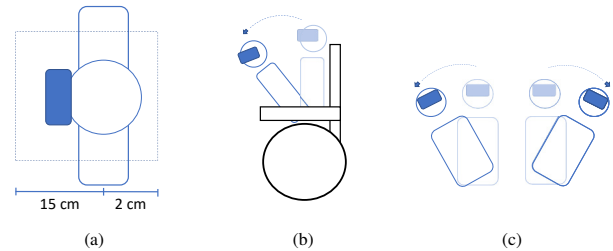


Figure 1: A typical leaning-based locomotion system. a) Upper view of the boundary the user must pass to start moving in the virtual environment (VE). b) Lateral view showcasing the forward and backward leaning movement. c) Front-view showing the lateral leaning that leads to a strafe movement in VE.

However, Gerling et al. [8] highlighted challenges such as limited physical space, the need for rotation, and complex setups associated with these systems. Addressing these limitations requires approaches that minimize physical demands, reduce reliance on rotational movements, and simplify setups, especially for users in constrained environments.

Building on these limitations, we propose a multimodal hands-free system tailored specifically for manual wheelchair users with upper limb mobility limitations. Using integrated HMD sensors, our system eliminates the need for additional hardware or complex configurations, directly addressing the spatial and setup challenges highlighted by Gerling et al. Our approach combines leaning-based translation [45] for intuitive forward and lateral navigation, head-gain rotation [19] for precise directional control, and gaze-based selection [37, 6] for efficient pointing and interaction. This integration ensures seamless interaction while minimizing physical effort, enabling users to navigate and interact within virtual environments more naturally. In addition, we explore refinements that improve the synergy between these modalities, ensuring a cohesive and adaptive experience that meets the unique needs of manual wheelchair users in constrained environments.

2 RELATED WORK

2.1 Wheelchair User Experience in Virtual Reality

In the context of accessibility, the design space in this domain can be described as either games that are tailored to individual abilities [44] or games that have been adapted to accommodate diverse abilities [9, 5]. Training and rehabilitation are some of the main applications that explored wheelchair use in VR. For instance, VR can be used to train specific motor skills through serious games [23], to simulate a wheelchair experience [33] or can leverage the wheelchair as input or output space [12]. Additionally, accessibility tools, such as embodied navigation systems [32] enable wheelchair users to navigate virtual replicas of real-world spaces through first-person avatar representations. Similarly, wheelchairs have also been used as input systems for robot navigation [39] or

for controlling movement in virtual environments [4, 11], such as via platforms that track wheelchair motion.

While handheld controls, body, and hand-tracking have been established as the default control systems in VR, researchers have explored several alternatives towards hands-free input. These include gestures [40], voice commands [17], and body movements using the torso [45] or the feet [3]. This variety of input has sparked interest in multimodal systems that address a wide range of 3D interactions: navigation, selection, and manipulation [20, 2]. Our proposal focuses on hands-free navigation and selection tasks for wheelchair users. A more in-depth discussion of the related work is provided in the following sections.

2.2 Hands-Free Locomotion for Wheelchair Users

Hands-free locomotion is typically supported via external mechanisms such as walk-in-place systems, stepper machines, or trackballs [3]. Pressure sensors embedded in a cushion have also been used to detect leg and body movements for direction and speed control in VR [31]. Similarly, wheelchair-based physical movements have enabled the exploration of virtual environments, particularly within room-scale settings [12, 3]. Techniques such as leaning or swaying, also called steering systems that rely on torso or head movements, effectively use the body as a human joystick towards locomotion [46, 30, 13] – this tends to assume free rotation of the user, such as sitting on swivel chairs. These approaches have shown a higher self-motion perception; especially relevant for exploratory tasks or continuous movements [14, 15]. However, limitations for wheelchair users appear in a dual task where they manually have to turn the wheelchair while leaning or performing tasks as selection or manipulation [8]. Therefore, systems that allow for 360° virtual rotation without the need for manual turns must be further explored.

Rotation gain systems address this limitation by amplifying virtual rotation relative to physical rotation [42, 34]. For instance, by having 90° of physical rotation produce a full 360° virtual rotation [19]. This rotation interaction can be completed with eye- or head-pointing systems, though to the best of our knowledge, they have not been combined with steering systems until now. Furthermore, systems that do not rely on body movements have also been explored, from using eye gazing for rotation [21] to voice controls for teleportation [17]. The latter shows particular benefits for long-distance exploration through landmarks or numbered grids in a virtual space [17]. While these approaches are effective for navigation, their integration with selection and manipulation tasks remains under-explored.

2.3 Hands-Free Pointing for Wheelchair Users

Eye gaze systems employing dwell time have shown comparative results to handheld controllers for object selection in VR [6, 28]. Smooth pursuit techniques have also been explored for similar purposes [7, 41], as well as systems combining head and gaze pointing [37]. Sidenmark et al. point out the benefit of separating pointing and confirmation tasks, particularly to select occluded objects [36] – multimodal systems can better support this separation of pointing and confirmation [29]. Similarly, voice commands can serve as an alternative input method, though a decrease in user performance has been observed when paired with eye-gaze systems [26]. Voice, head-gaze, and eye-gaze interfaces are among the most extensively studied hands-free systems for selection tasks [27]. In the context of wheelchair navigation, gaze-based systems have been positioned as intuitive, hands-free solutions that align well with users' physical constraints.

3 DESIGN AND IMPLEMENTATION

Leaning-based navigation, combined with gaze-based selection, offers a promising hands-free interaction framework for multimodal VR interaction [45]. However, existing implementations re-

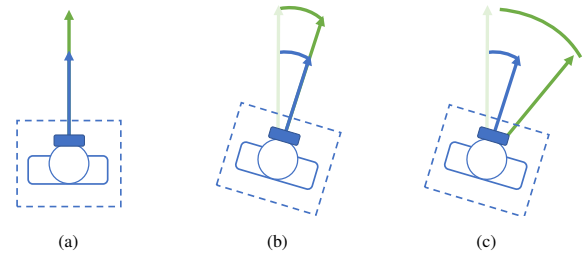


Figure 2: Our camera rotation system, with the dotted boundary indicating the limits defined by the locomotion system. a) The initial state (null direction) where the physical (blue arrow) and virtual (green arrow) rotations are aligned. b) The start of a camera rotation, with the user's head still in sync with the virtual rotation. c) The physical rotation remains in the same position (blue arrow), while the camera continues to rotate to the right at a constant angular speed (green arrow). The user must face forward again (null direction) to stop the camera from continuing to rotate.

quire users to physically rotate, a known limitation for wheelchair users if performing multiple tasks. Additionally, there is a trade-off between navigation and selection efficiency in virtual environments [24]. In terms of selection mechanisms, raycasting enabled faster task completion than virtual hands and particularly used in short distance ranges [24]. Therefore, we propose a leaning-based navigation system incorporating dynamic head gains for 360° virtual rotations. This system is paired with gaze-based raycast pointing (using either eye or head tracking) to enhance pointing tasks. These components aim to create a cohesive, hands-free interaction for wheelchair users in VR.

3.1 Leaning-based Locomotion

Similarly to previous studies [47], users navigate a 3D environment by leaning forwards or backwards, left or right (strafing), or via a combination of the two. In our implementation, we use head-tracking data from the VR headset, derived from its accelerometer and gyroscope, to extract relative position and rotation. The velocity of virtual movement has a linear relation to the leaning angle, measured by the displacement of the head position relative to a null point registered at the start of the session.

Fig. 1 illustrates how leaning movements interact with the locomotion system. Fig. 1a highlights the boundaries of the dead-zone, within which no locomotion occurs. These boundaries are determined by the torso's range of motion, as described by Lura [22]. The backward boundary is positioned closer to the null point since users cannot lean back (25°) as much as they can lean forward (80°) – see Fig. 1b. A seated posture further influences this constraint. The lateral leaning boundary shown in Fig. 1c is also defined as close to the null point to reflect the 35° range of lateral motion reported by Lura [22]. These movements can occur in tandem, allowing users to effectively leverage their torsos as a joystick.

3.2 Dynamic Head-Gain Rotation

The basic principle of amplified rotation applies a constant amplification factor to the real head rotation angle, enabling up to 180° of virtual rotation on either side. For instance, a 2X amplification factor means that a physical angle rotation of 60° (θ_{real}) represents a 120° virtual angle rotation ($\theta_{virtual}$). Dynamic amplification mechanisms offer greater flexibility by adjusting their amplification based on factors such as time, speed, or via non-linear relations, allowing for smoother transitions within a comfortable range and enabling up to 360° virtual rotations [19].

We adopted a dynamic amplification system for rotation [19],

which continuously increases the virtual rotation based on the user's physical rotation angle. This approach aims to provide users with greater freedom for continuous virtual exploration, while making the interaction independent from secondary devices. Additionally, we assume that the physical forward direction is fixed, as shown in Fig. 2a, so that the virtual rotation stops when users return to this initial forward position. To provide a smooth experience, we introduced a slight delay in the virtual rotation relative to the physical rotation, using a spherical interpolation technique for a smoother ease-in and ease-out transition. As a result, when the user reaches the desired physical angle (θ_{real}) and stops (Fig. 2b), the virtual rotation continues in the same direction (Fig. 2c), conserving its angular speed until the user changes their physical angle (θ_{real}). This behaviour, illustrated in Fig. 2, demonstrates the relationship derived from spherical interpolation where ω corresponds to the angular speed: $\omega_{virtual}(t) \approx 2.5 \times \omega_{real}$. The 2.5 amplification factor was selected as an approximate mid-value based on the findings of Zhang et al. [42].

This continuous system is designed to work along with the leaning-based locomotion implementation described above. Based on Lee et al.'s system [21], we consider that adding a zone that allows users to freely move without causing the camera to rotate could decrease the chances of cybersickness – we describe this as the dead-zone. In future work, this dead-zone could be further explored to support a gazing mechanism that serves both rotation and pointing purposes.

3.3 Gaze Pointing System

Considering the work from Zielasko et al. [46], we conclude our implementation by adding head-gaze pointing to the navigation system described so far (i.e., locomotion and dynamic camera rotation). This describes a raycast from the centre of the user's field of view (FoV) that acts as an (invisible) pointer [24]. In sum, users lean their torso to move in 3D space and their heads for both rotation and pointing tasks. Our approach only requires head-tracking data from the gyroscopes easily accessible from most VR headsets. Since using the head for two purposes can potentially lead to unwarranted exertion, we also implemented a version of our pointing system using a less available input modality, eye-gaze – freeing the head for sole camera control.

A visual summary of our system can be seen in Fig. 3, in which we propose the use of leaning-based locomotion, a continuous dynamic head gain for camera rotation, and the use of either head- or eye-gaze for pointing purposes.

4 DISCUSSION

A key consideration in this context is understanding the spectrum of disabilities among wheelchair users and how this proposal fits their needs. We identify the main physical abilities necessary to use the system. This proposal requires a certain degree of torso movement for leaning and head rotation for interaction, as well as it assumes free or corrected vision from at least one eye in the case of the eye-gaze pointing.

The system we presented in this paper was designed for wheelchair users with upper body limitations, i.e., users who cannot operate hand-held controllers. To achieve this we had to consider how to support three seminal activities in a hands-free manner: locomotion in a 3D scene, camera rotation, and pointing. While, by themselves, these are not new techniques, we believe this is the first time they have been considered and implemented together. For instance, for systems using leaning-based locomotion, the camera rotation was achieved by turning a swivel chair around [30] – something a wheelchair user might struggle to do while immersed in a VR experience in its home. Not only that, but we believe our approach could be useful to non-wheelchair users with upper limb limitations as well. Many users still engage with VR experiences

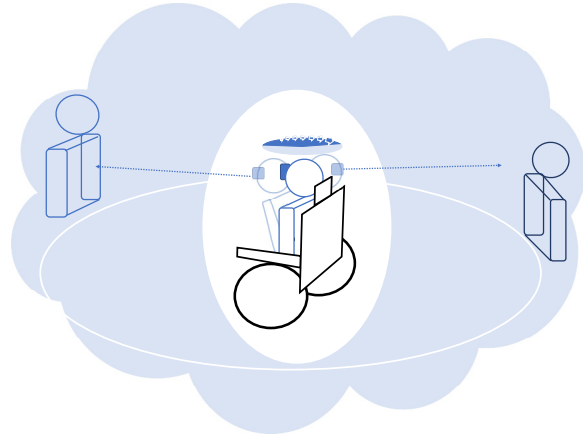


Figure 3: A summary of our multimodal and hands-free input system for wheelchair users. Leaning is used for locomotion in a virtual environment, the head controls camera rotations in 360° without having to rotate the wheelchair, and users can rely on head- or eye-gaze for pointing at areas of interest.

while sitting on a couch or at their desks [38] and thus are unwilling to spin around.

We will evaluate the pointing methods (eye-gazing versus head-gazing) and the integration of a dead-zone in the rotation mechanism to mitigate cybersickness. Furthermore, we aim to assess the overall usability, comfort, and effectiveness of the system. This includes evaluating its ease of use, user satisfaction, and the ability to provide a comfortable and efficient experience.

5 CONCLUSION

In this paper we proposed integrating three specific systems to enable a hands-free experience of 3D environments for manual wheelchair users with upper limb limitations without specialized equipment other than a VR headset. Our approach aims to widen these users' access to popular VR experiences ranging from games to the metaverse, in which avatar movement and rich interactions are expected. Our system supports locomotion, rotation, and pointing through leaning, head-gain rotation, and head- or eye-gaze, respectively. Future work should establish not only the user experience of our approach (e.g., user performance, physical effort, cybersickness), but optimize some of our design choices (e.g., dead-zone, head- vs eye-gaze pointing).

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